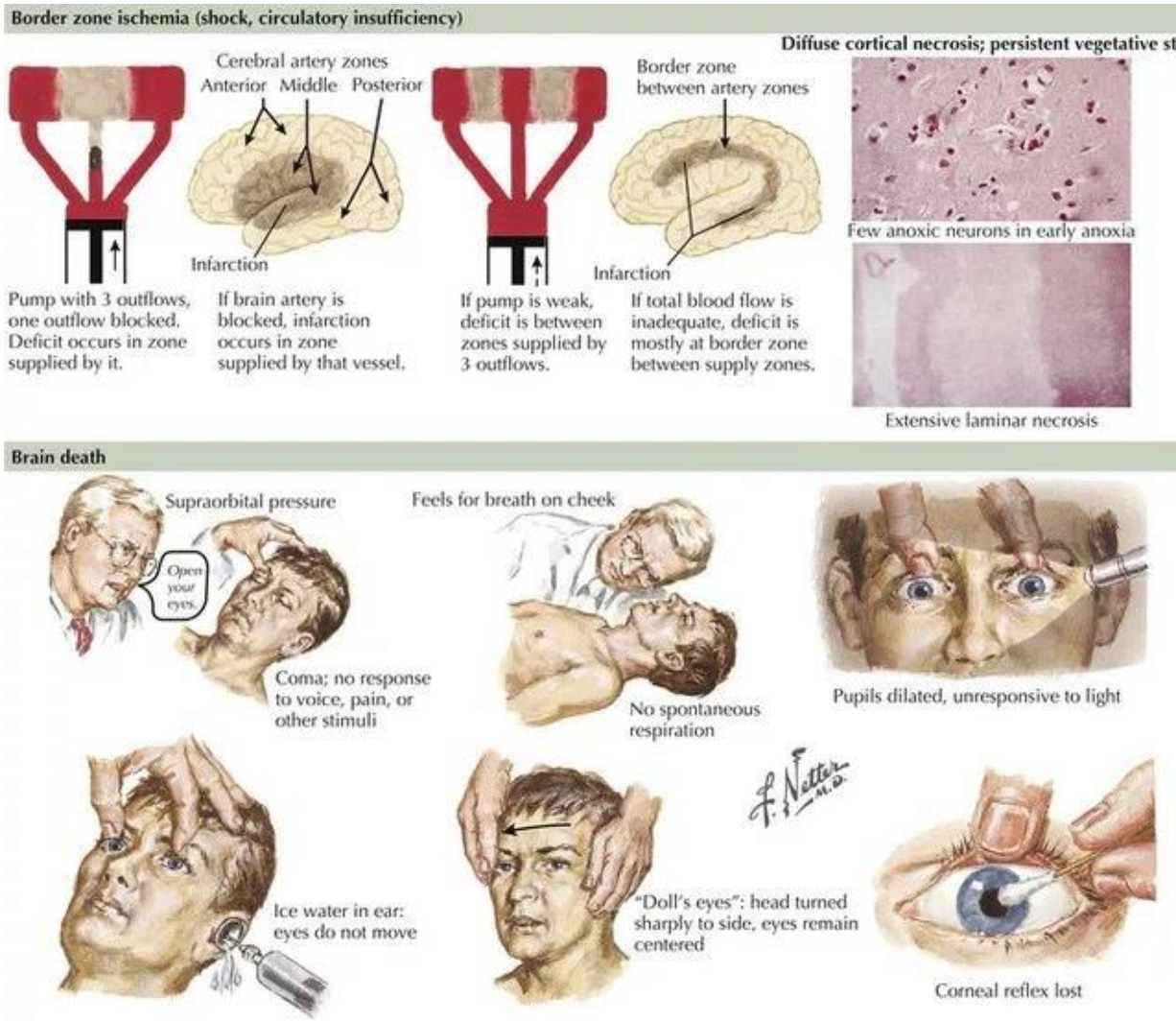
Death - Recognition and Certification Policy



©Hope Hospital

Recognition of death

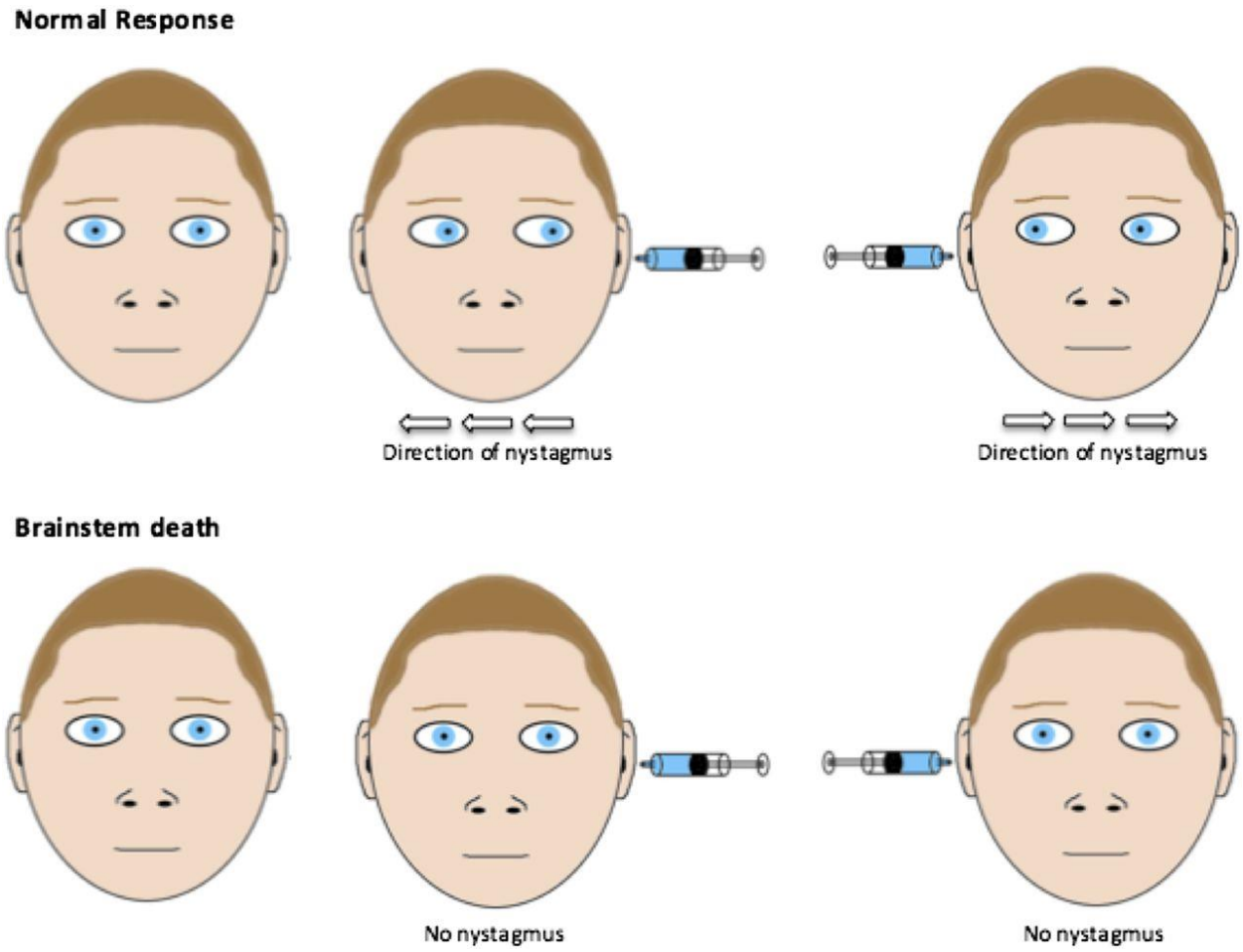
It is vital when certifying death, to ensure that death has indeed occurred. In the modern world of advanced intensive care techniques and potential for organ donation, this can be a challenge. Death is the irreversible loss of capacity for consciousness combined with the irreversible loss of capacity to breathe.

LEGALLY ACCEPTED CRITERIA FOR BRAIN DEATH

- Brain death is a loss of brainstem reflexes such as:
- **Pupillary light reflex**
- **Corneal reflexes**
- **Doll's eyes (oculocephalic) reflexes**
- **Caloric responses to iced water stimulation of the tympanic membrane**
- **Absence of spontaneous respiration**

It is an irreversible loss of functioning in the entire brain, both cortex and brainstem.

©Hope Hospital



Guidance on the diagnosis and confirmation of death

Proceed without unnecessary and distressing delay. Death may be obvious with clear signs pathognomonic of death (hypostasis, rigor mortis). If not, obvious death should be identified by "the simultaneous and irreversible onset of apnoea and unconsciousness in the absence of the circulation".

The guidance in addition advises that:Full and extensive attempts at reversal of any contributing cause to the cardiorespiratory arrest have been made where appropriate (for example, body temperature, endocrine, metabolic and biochemical abnormalities more relevant in hospital).

- One of the following is fulfilled:

    - The individual meets the criteria for not attempting cardiopulmonary resuscitation
    - Attempts at cardiopulmonary resuscitation have failed
    - Treatment aimed at sustaining life has been withdrawn because it has been decided to be of no further benefit to the patient and not in his/her best interest to continue and/or is in respect of the patient's wishes in an advance decision.
- The individual should be observed by the person responsible for confirming death, for a minimum of five minutes to establish that irreversible cardiorespiratory arrest has occurred. In primary care the absence of mechanical cardiac function is normally confirmed using a combination of the following:
    - Absence of a central pulse on palpation.
    - Absence of heart sounds on auscultation.
- In hospital this can be supplemented by one or more of the following:
    - Asystole on a continuous ECG display.
    - Absence of pulsatile flow using direct intra-arterial pressure monitoring.
    - Absence of contractile activity using echocardiography.
- Any spontaneous return of cardiac or respiratory activity during this period of observation should prompt a further five minutes of observation from the next point of cardiorespiratory arrest.
- After five minutes of continued cardiorespiratory arrest the absence of the pupillary responses to light, of the corneal reflexes, and of any motor response to supraorbital pressure should be confirmed
- The time of death is recorded as the time at which these criteria are fulfilled.

Verifying death

Who can verify death?

- Does not require a doctor to confirm death has occurred or that "life is extinct".
- Does not require a doctor to view the body of a deceased person.
- Does not require a doctor to report the fact that death has occurred.
- Does require the doctor who attended the deceased during the last illness to issue a certificate detailing the cause of death (unless the death is referred for postmortem).

So a doctor's legal duty is to notify the cause of death, not the fact that death has taken place. Doctors, nurses or suitably trained ambulance clinicians may confirm that death has taken

©Hope Hospital

place. There is no legal obligation on a doctor to see or examine the deceased before signing a death certificate.

Should a GP visit? For deaths in the community, the policy advises the following:

Expected deaths:

- Death in the patient's own home: visit as soon as possible, whilst prioritising the urgent needs of living patients.
- Death in a residential or nursing home: If the GP who attended the patient during the last illness is available, he/she should attend when practicable and issue a death certificate. If that GP is unavailable, it is unlikely that any useful purpose will be served by a duty doctor attending. The duty GP may advise the home to contact the undertaker if they wish the body to be removed and notify the GP with whom the patient was registered as soon as practicable.

Unexpected deaths:

- A visit by the GP with whom the patient was registered, to examine the body and confirm death, although this is not a statutory requirement.
- If the unexpected death occurs out-of-hours it is helpful if a duty GP does attend, as this may prevent the potentially unnecessary attendance of the emergency services.

How to verify death

A thorough physical examination should be carried out to ascertain whether or not death has taken place. First inspection should reveal an extreme pallor (particularly of the face and lips) and relaxation of the facial muscles. This leads to drooping of the lower jaw and open staring eyes, unless these have been closed. Further examination should confirm:

- No palpable pulses.
- No heart sounds on auscultation (or asystole on ECG).
- No observed respiratory effort.
- No breath sounds on auscultation.
- Pupils dilated and not reactive to light.

Particularly if the death is unexpected, an external examination of the deceased and their surroundings should be made, to look for any apparent factors which may be relevant to their death (bleeding, vomit, wounds, weapons, alcohol, pills, notes, etc).

Other signs include:

©Hope Hospital

- No response to painful stimuli.
- Absence of corneal reflexes.
- Cloudiness of the cornea.
- Examination of the trunk may show evidence of post-mortem staining as a result of hypostasis.
- Rigor mortis may have set in (begins approximately three hours after death).
- Decreased temperature - will depend on ambient temperature but may not occur for up to eight hours.

The precise moment of death may be difficult to recognise, and for a period of time after respiration has ceased, and the heart has stopped, the patient may still potentially be resuscitated. In certain conditions a patient may appear dead if not thoroughly examined:

- Following prolonged submersion in cold water.
- Following ingestion of alcohol or drugs.
- When hypoglycaemic or in a coma.

They may recover completely, if treated appropriately. It should be remembered that hypothermia protects against hypoxic neurological damage and that children under the age of 5 are more resilient to hypoxic brain injury; therefore, resuscitation should be continued in these circumstances until normal body temperature is reached, even if the patient appears to be dead.

Practical definition of death in primary care

For practical purposes in General Practice, death is usually deemed to exist in an unresponsive patient, with a body temperature over 35°C, who has not been taking drugs or alcohol if:

- There are no spontaneous movements.
- There is no respiratory effort (examine for at least one minute).
- There are no heart sounds or palpable pulses (examine for at least one minute).
- There is an absence of reflexes - eg, corneal.
- The pupils are fixed and dilated.

Certification of death

©Hope Hospital

Management of a death will depend on:

- The circumstances of the death.
- Where it has occurred.
- Whether or not it was anticipated.
- Whether or not there is any suspicion of foul play.

Relatives and/or friends of the deceased may be very distressed and GPs attending a death should offer support where appropriate. Bereaved families may also require guidance on the procedures following a death, particularly if the death was unexpected.

Medical certificate of cause of death (MCCD)

The death certificate fulfils a number of purposes:

- It allows the relatives of the deceased to register the death.
- It provides a permanent legal record of death
- It allows the relatives to arrange for the funeral, etc and to settle the estate of the deceased.
- It is used to provide national statistics about causes of death and trends in disease which go on to guide research, health services planning, etc.

A death certificate may be issued by a doctor who has provided care during the last illness and who has seen the deceased within 14 days of death or after death. They should be confident about the cause of death.

In some circumstances, a doctor is unable to provide a death certificate and the death must be reported to the Police as a MLC. Such circumstances include:

- No doctor satisfies the attendance requirements for being able to certify death - eg, the only doctor who has provided care during the last illness is away on holiday, or the deceased has not been seen by a doctor within the preceding 14 days.
- If the cause of the death is unknown.

©Hope Hospital

- Sudden, unexpected, suspicious, violent (homicide, suicide, accidental) or unnatural deaths.
- Deaths resulting from injury or poisoning.
- Deaths due to alcohol or drugs. (Not chronic alcohol or tobacco use.)
- Doubtful stillbirth.
- Deaths related to surgery or anaesthetic.
- Deaths within 24 hours of admission to hospital.
- Deaths in prison.
- Identity of deceased unknown.
- Death from an industrial disease.
- Death from neglect, want or exposure.

A few specific points are worth mentioning regarding filling of death certificate::

- Old age. Old age alone should not be used as the sole cause on a death certificate unless:
    - The deceased is 80 years of age or more.
    - You have personally cared for the deceased over a long period (years or many months).
    - You have observed a gradual decline in your patient's general health and functioning.
    - You are not aware of any identifiable disease or injury that contributed to the death.
    - You are certain that there is no reason that the death should be reported to the coroner.
    - You have considered checking with relatives that they are satisfied with this explanation for the cause of death.
- Organ failure. Avoid organ failure alone as the cause of death. Specify the condition which led to organ failure below.
- Mode of dying or terminal events. These cannot be used as the cause of death (eg, cardiac arrest or shock).
- Abbreviations. Do not use abbreviations on a death certificate.
- Diabetes. Specify type 1 or type 2 and give the complication which led to death.

©Hope Hospital

The death certificate is given to the next of kin who is required to deliver it to the Registrar of Births, Deaths and Marriages within five days. In the absence of a next of kin, the following can register the death:

- A relative.
- Someone present at the death.
- An administrator from the hospital.

Summary: what to do when called to a death in primary care

- Be sensitive and supportive towards bereaved and/or shocked relatives.
- Verify that death has taken place as in the section above.
- Document the time you verified death.
- In an expected death, provide a death certificate as soon as possible.
- In an unexpected death, document anything on or around the body which may point towards a cause of death. Explain to the relatives it may not be possible to issue a certificate until a cause of death has been established, and to do this you will need to refer the death for Postmortem.
- Where relevant, follow up with ongoing offers of support to the family.

As a junior doctor, you will often be called to the ward to confirm death in a patient. It is first important to recognise that dying is a process leading to death. Where a delay to the confirmation of death exists, such as in the pre-hospital or primary care setting, there is often no doubt that the patient has died.

However, in a hospital setting, where such a delay should not exist, the distinction is less clear. We therefore use the guidelines, stating that death has occurred when:

- There has been a simultaneous onset of apnoea and unconsciousness in the **absence of circulation** that is irreversible…
- … following **extensive attempts** to treat any contributing factors that may have led to cardiorespiratory arrest.

©Hope Hospital

The second point described here is not necessary where a valid **'Do Not Attempt Resuscitation' order** is in place. It is also not valid in a situation where treatment aimed at sustaining life has been withdrawn because it had been decided to inappropriate (i.e. of no further benefit to the patient and not in their best interests) or at the patient's request.

To confirm 'brainstem death' in a patient (whereby cardiorespiratory function is maintained by artificial means but where the irreversible cessation of their brainstem function has occurred) see the policy on brainstem death.

The Final Examination

When you have been asked to confirm the death of a patient, you should observe the patient for a minimum of 5 minutes. To ensure that you perform all necessary steps, you can use a systematic A to E approach:

- **Airway / Breathing** – Auscultate the lungs for >1 min
    - There will be no respiratory effort and no audible breath sounds.
- **Circulation** – Palpate for a pulse for >1 min and auscultate the heart for >1 min
    - There will be no palpable central pulse and no audible heart sounds*
    - Make sure you check for any palpable cardiac pacemaker
- **Disability** – Check for a pupillary response and check for a motor response to pain
    - Following 5 minutes of continued cardio-respiratory arrest the patient's pupils will be fixed, dilated and unresponsive to light.
    - There will be no response to a painful stimulus. This can be tested by applying supra-orbital pressure and looking for any motor response.
- **Exposure** – The patient may be peripherally cold (depending on the timing of your assessment)

**In a patient who had been monitored, you may also notice continuous asystole on the cardiac monitor. In a patient who has an arterial line you can observe an absence of pulsatile flow.*

Documentation

The above examination will often be documented in the notes as below:

DATE 02/12, TIME: 2310: DEATH CERTIFICATION

No audible breath or heart sounds for greater than 1 minute.

No palpable pulse for greater than 1 minute. No palpable cardiac pacemaker.

Pupils are fixed, dilated and unreactive to light.

There is no response to painful stimulus.

©Hope Hospital

Asystole on a continuous ECG display./ Absence of pulsatile flow using direct intra-arterial pressure monitoring./Absence of contractile activity using echocardiography.

The patient has died.

During the process of confirmation of death, the duty doctor will need to make yourself available to discuss any issues around the circumstances of death or hospital admission with the patient's next of kin. If the family are not present at the time of death, it is the duty doctor's duty to ensure that they are informed immediately.

©Hope Hospital